





CIRP - Literature Review

Customer Intelligence & Retention Platform

DEPI • AI & Machine Learning Track • Round 5

Graduation Project

Project	Customer Churn Prediction & Segmentation Platform (CIRP)
Track	DEPI - AI & Machine Learning
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1. Introduction

This review covers how churn is defined in retail, which methods are used for churn prediction and segmentation, and how they shape the design of CIRP.

2. Customer Churn in Retail

In retail the customer relationship is non-contractual, so churn needs an operational definition: no purchase within a fixed window after an observation date. CIRP labels a customer as churned if there is no order in the 90 days after the snapshot date.

The data is Online Retail II (a UK online retailer, 2009-2011). Many customers are wholesalers, so orders are large and irregular and most features are strongly skewed.

3. Machine Learning for Churn Prediction

3.1 Algorithms

Churn prediction is a binary classification problem. Comparative studies test the same classifier families and show that richer behavioral features improve results.

- **Logistic Regression:** fast and explainable through its coefficients; the standard baseline.
- **Decision Tree:** readable rules, but can overfit.
- **Support Vector Machine:** effective on scaled numeric features, but harder to explain.
- **Random Forest and Gradient Boosting:** tree ensembles that often give the best accuracy on tabular data.

CIRP compares Logistic Regression, Decision Tree and Random Forest; the tuned Random Forest was kept as the final churn model.

3.2 Evaluation and class imbalance

Accuracy hides poor results here: 61% of the 22,372 customer-snapshots are labelled churn, so the majority class alone already reaches 61% accuracy and accuracy does not show whether the model finds the right customers. The precision-recall curve is more informative than ROC under this imbalance, and Precision, Recall and F1 separate the two errors. Missing a churner costs more than an unnecessary offer, so CIRP reports Precision, Recall, F1, ROC-AUC and PR-AUC and chooses the threshold on validation data: for the Random Forest, lowering the threshold from 0.50 to 0.45 raised Recall from 0.914 to 0.943 and lowered Precision from 0.800 to 0.785, and 0.45 was kept.

3.3 Explainability, feature selection and leakage

Coefficients and tree feature importances show which factors drive churn; CIRP uses global feature importance.

CIRP selects features on training data only: correlation with the target, removal of pairs above 0.85 and an F-test ranking. This removed 3 of the 22 features (orders_last_30d, orders_last_60d, spend_trend)

and left 19, and the strongest single correlation with churn is 0.34, so the model relies on the combination of features rather than on one signal. Leakage gives over-optimistic results, so CIRP splits by time and uses the test period once.

4. Customer Segmentation

4.1 RFM features

RFM (Recency, Frequency, Monetary value) combined with K-Means is the classic approach to retail segmentation and produces actionable customer groups. CIRP adds average order value, product variety, return rate and spending trend.

4.2 Clustering algorithms

- **K-Means:** fast and interpretable, but assumes compact groups and is sensitive to scale and outliers.
- **DBSCAN:** finds dense regions and marks outliers as noise without a fixed k, but is sensitive to its parameters.

RFM variables are skewed, so CIRP applies a log transform and scaling, then compares K-Means with DBSCAN.

4.3 Choosing and validating clusters

Without labels, clusters are judged by the Silhouette coefficient, the Davies-Bouldin index and the elbow plot. CIRP also profiles each segment (mean RFM, size, churn rate) and gives it a business name and a suggested action.

5. Research Gap and Positioning of CIRP

Earlier work on e-commerce customers segmented them with K-Means and then predicted churn, reporting better results after segmentation. CIRP tests this by comparing the churn model with and without the segment as a feature. Most reviewed work uses contract-based telecom data; CIRP uses non-contractual retail data and combines both models in one reproducible application.

6. Lecturer Feedback & Evaluation

Feedback & Evaluation and Final Grading Criteria are reserved for the lecturer.

Feedback & Evaluation

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Suggested Improvements

- **Value-based targeting:** rank at-risk customers by churn probability times expected spend, so the retention budget goes first to customers who are both likely to leave and worth keeping.
- **Customer-specific churn window:** wholesale customers order irregularly, so a fixed 90-day label can mark a normal gap as churn; define churn relative to each customer's own typical time between orders.
- **Segment migration tracking:** use the 5 snapshots to see how customers move between segments and treat a move to a weaker segment as an early warning signal.
- **Reasons per prediction:** show the top factors behind each customer's churn risk in the app, so the business sees why and can choose the right action.
- **New customers:** customers with a single order behave differently from repeat buyers; handle them with a separate rule or model instead of the same one.

Final Grading Criteria

Criterion	Maximum marks	Marks awarded
Documentation		
Implementation		
Testing		
Presentation		
Total		